Daniel L. Goldberg, Ph.D.

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Assistant Research Professor
Milken Institute School of Public Health
Department of Environmental and Occupational Health
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Professional Appointments

2021–present	Assistant Research Professor	George Washington University
2019–2021	Staff Research Scientist	George Washington University
2019–2021	Staff Research Scientist	Argonne National Laboratory
2016–2019	Postdoctoral Scientist	Argonne National Laboratory

Academic Degrees

2015	Ph.D.	University of Maryland, Atmospheric & Oceanic Science
2013	M.S.	University of Maryland, Atmospheric & Oceanic Science
2009	B.S.	Lafayette College, Chemical Engineering

Scientific and Technical Background

My research uses satellite data and models to characterize global air pollution and greenhouse gas emissions at neighborhood scales. My work primarily revolves around two satellite instruments: the TROPOspheric Monitoring Instrument (TROPOMI) and the Tropospheric Emissions Monitoring of POllution (TEMPO). Both instruments measure a toxic air pollutant, nitrogen dioxide (NO_2) that results primarily from fossil fuel combustion. NO_2 is closely linked to other atmospheric pollutants, such as ozone (O_3), particulate matter ($PM_{2.5}$), and carbon dioxide (CO_2). Results from his research have identified sources of air pollution and greenhouse gas emissions that have been previously mischaracterized or unknown, ultimately leading to better local, state, federal, and international policies and guidance.

Peer-Reviewed Publications (17 first-author, 3 senior-author, 58 total; h-index: 30)

2026 (1 first-author, 4 total)

1. **Goldberg, D. L.**, Johnson, J., Nawaz, M. O., Kerr G. H., Chen, L., Huber, D., Judd, L. M. Contrasting the TEMPO and TROPOMI Views of Tropospheric NO_2: Implications for Diurnal Variability and Model Evaluation, *Journal of Geophysical Research: Atmospheres*, https://doi.org/10.1029/2026JD046905, **2026**.
2. Huber, D. E., Kerr, G. H., Nawaz, M. O., Runkel, S., Anenberg, S. C., **Goldberg, D. L.** Global NO_2 changes between 2019 and 2024 as observed by TROPOMI in urban areas and emerging hotspots, *Atmospheric Chemistry & Physics,* https://doi.org/10.5194/acp-26-3783-2026, **2026**.
3. Jaffe, D., Lee, H. B., **Goldberg, D. L.**, O'Dell, K., Magzaman, S. Health and Regulatory Impacts of PM2.5 from Wildland Fires for 2019-2024 in the U.S. *GeoHealth,* https://doi.org/10.1029/2025GH001576, **2026**.
4. Eckel, S. P., Chen, F., Silva, S. J., **Goldberg, D. L.**, Johnston, J., Palinkas, L. A., Campos, A., Franco, W., and Garcia, E.: Zero-emissions vehicle adoption and satellite-measured NO2 air pollution in California, USA, from 2019 to 2023: a longitudinal observational

study, *Lancet Planet. Health*, 101379, https://doi.org/10.1016/j.lanplh.2025.101379, **2026**.

<u>2025 (1 first-author, 9 total)</u>

5. **Goldberg, D. L.,** Nawaz, M. O., Lyu, C., He, J., Carlton, A. G., Kondragunta, S., and Anenberg, S. C.: Clear-sky and cloudy-sky differences in NO_2 concentrations over the United States: implications for satellite measurement applications, *Atmos Chem Phys*, 25, 16287–16302, https://doi.org/10.5194/acp-25-16287-2025, **2025.**
6. Siu, T. K., Goldberg, D. L., Kerr, G. H., Chen, L., Nawaz, M. O., Chang, R. Y. -W., and Fong, K. C.: Tropospheric NO2 Patterns in Eastern Canada Using the First Year of TEMPO Observations, *Journal of Geophysical Research: Atmospheres*, 130, https://doi.org/10.1029/2025JD044757, **2025**.
7. Burkart, K., Wozniak, S., Anenberg, S., Pereda, A., Gilbertson, N., Ashbaugh, C., **Goldberg, D.**, Hystad, P., Kerr, G. H., McLaughlin, S. A., Mohegh, A., and Brauer, M.: Global, regional and national estimates of the burden of childhood asthma attributable to NO2 exposure for 204 countries and territories from 1990 to 2023: a Global Burden of Disease study 2023, *EClinicalMedicine*, 90, 103580, https://doi.org/10.1016/j.eclinm.2025.103580, **2025**.
8. Nawaz, M. O., Huber, D. E., Kerr, G. H., Judd, L. M., Acker, S. J., and **Goldberg, D. L.**: A Comparative Analysis of TEMPO NO_2 Remote Sensing With Surface-Level Monitoring Through Diurnal and Seasonal Trends, Meteorology, and Monitor Characteristics, *Journal of Geophysical Research: Atmospheres*, 130, https://doi.org/10.1029/2025JD043923, **2025**.
9. Farzad, K., Zhang, Y., Wang, K., Chen, X., **Goldberg, D. L.**, Lyapustin, A., Wang, Y., and Bell, M. L. Statistical downscaling of coarse-resolution 3-D model $PM_{2.5}$ predictions to 1-km over the Contiguous United States: Model development, evaluation, and implication in health impact assessment, *Science of the Total Environment,* https://doi.org/10.1016/j.scitotenv.2025.180302, **2025**.
10. Ahn, D. Y., **Goldberg, D. L.,** Liu, F., Anderson, D. C., Coombes, T., Loughner, C. P., Kiel, M., Chatterjee, A. Regional and Socioeconomic Characteristics in C40 Cities' CO_2 Emissions Revealed from Space, *AGU Advances*, https://doi.org/10.1029/2025AV001747, **2025**.
11. Lawal, A., Skipper, T.N., Ivey, C. E., **Goldberg, D. L.,** Kaiser, J., Russell, A. G. Potential Errors in CMAQ NO:NO_2 Ratios and Upper Tropospheric NO_2 Impacting the Interpretation of TROPOMI Retrievals, *Environmental Science & Technology – Air,* https://pubs.acs.org/doi/full/10.1021/acsestair.4c00198, **2025**.
12. Nawaz, M. O., **Goldberg, D. L.,** Kerr, G. H., and Anenberg, S. C.: Incorporating fine resolution observations from TROPOMI reduces complexity and improves predictive accuracy of LUR modeling of NO_2 in the US, *Environmental Science & Technology – Air,* https://pubs.acs.org/doi/10.1021/acsestair.4c00153, **2025**.

<u>2024 (2 first-author, 7 total)</u>

13. **Goldberg, D. L.**, B. de Foy, M. O. Nawaz, J. Johnson, G. Yarwood, L. Judd. Quantifying NO_x Emission Sources in Houston, Texas Using Remote Sensing Aircraft Measurements

and Source Apportionment Regression Models, *Environmental Science & Technology – Air*, https://pubs.acs.org/doi/10.1021/acsestair.4c00097, **2024**.

14. **Goldberg, D. L.,** Tao, M., Kerr, G. H., Ma, S., Tong, D. Q., Fiore, A. M., Dickens, A. F., Adelman, Z. E., and Anenberg, S. C.: Evaluating the spatial patterns of U.S. urban NOx emissions using TROPOMI NO_2, *Remote Sens Environ*, 300, 113917, https://doi.org/10.1016/j.rse.2023.113917, **2024**.
15. Kim, E. J., Holloway, T., Kokandakar, A., Harkey, M., Elkins, S., **Goldberg, D. L.,** and Heck, C.: A Comparison of Regression Methods for Inferring Near-Surface NO_2 With Satellite Data, *Journal of Geophysical Research: Atmospheres*, 129, https://doi.org/10.1029/2024JD040906, **2024**.
16. Kerr, G. H., Meyer, M., **Goldberg, D. L.,** Miller, J., and Anenberg, S. C.: Air pollution impacts from warehousing in the United States uncovered with satellite data, *Nature Communications*, 15, 6006, https://doi.org/10.1038/s41467-024-50000-0, **2024**.
17. Nawaz, M. O., Johnson, J., Yarwood, G., de Foy, B., Judd, L., and **Goldberg, D. L.:** An intercomparison of satellite, airborne, and ground-level observations with WRF–CAMx simulations of NO2 columns over Houston, Texas, during the September 2021 TRACER-AQ campaign, *Atmos Chem Phys,* 24, 6719–6741, https://doi.org/10.5194/acp-24-6719-2024, **2024**.
18. Kerr, G. H., van Donkelaar, A., Martin, R. V., Brauer, M., Bukart, K., Wozniak, S., **Goldberg, D. L.,** and Anenberg, S. C.: Increasing Racial and Ethnic Disparities in Ambient Air Pollution-Attributable Morbidity and Mortality in the United States, *Environ Health Perspectives*, 132, https://doi.org/10.1289/EHP11900, **2024**.
19. O'Dell, K., Kondragunta, S., Zhang, H., **Goldberg, D. L.,** Kerr, G. H., Wei, Z., Henderson, B. H., and Anenberg, S. C.: Public Health Benefits From Improved Identification of Severe Air Pollution Events With Geostationary Satellite Data, *GeoHealth*, 8, https://doi.org/10.1029/2023GH000890, **2024**.

<u>2023 (0 first-author, 6 total)</u>

20. Kerr, G. H., **Goldberg, D. L.**, Harris, M. H., Henderson, B. H., Hystad, P., Roy, A., and Anenberg, S. C.: Ethnoracial Disparities in Nitrogen Dioxide Pollution in the United States: Comparing Data Sets from Satellites, Models, and Monitors, *Environ Sci Technol*, https://doi.org/10.1021/acs.est.3c03999, **2023**.
21. Tzortziou, M., Loughner, C. P., **Goldberg, D. L.**, Judd, L., Nauth, D., Kwong, C. F., Lin, T., Cede, A., and Abuhassan, N.: Intimately tracking NO_2 pollution over the NYC - Long Island Sound land-water continuum: An integration of shipboard, airborne, satellite observations, and models, *Science of The Total Environment*, 165144, https://doi.org/10.1016/j.scitotenv.2023.165144, **2023**.
22. Larkin, A., Anenberg, S., **Goldberg, D. L.,** Mohegh, A., Brauer, M., and Hystad, P.: A global spatial-temporal land use regression model for nitrogen dioxide air pollution, *Front Environ Sci*, 11, 484, https://doi.org/10.3389/FENVS.2023.1125979, **2023**.
23. Ahn, D. Y., **Goldberg, D. L.,** Coombes, T., Kleiman, G., Anenberg, S. C.: CO_2 Emissions from C40 Cities: Citywide Emission Inventories and Comparisons with Global Gridded Emission Datasets, *Environmental Research Letters*, 18, 034032, https://doi.org/10.1088/1748-9326/acbb91, **2023**.

24. Nawaz, M. O., Henze, D. K., Anenberg, S. C., Ahn, D. Y., **Goldberg, D. L.,** Tessum, C. W., and Chafe, Z. A.: Sources of air pollution-related health impacts and benefits of radially applied transportation policies in 14 US cities, Frontiers in Sustainable Cities, 5, https://doi.org/10.3389/frsc.2023.1102493, **2023**.
25. Bechle, M. J., Bell, M. L., **Goldberg, D. L.,** Hankey, S., Lu, T., Presto, A. A., Robinson, A. L., Schwartz, J., Shi, L., Zhang, Y., and Marshall, J. D.: Intercomparison of Six National Empirical Models for PM2.5 Air Pollution in the Contiguous US, *Findings*, https://doi.org/10.32866/001c.89423, **2023**.

2022 (1 first-author, 6 total)

26. **Goldberg, D. L.,** Harkey, M., de Foy, B., Judd, L., Johnson, J., Yarwood, G. and Holloway, T.: Evaluating NOx emissions and their effect on O3 production in Texas using TROPOMI NO2 and HCHO, Atmos. Chem. Phys., 22(16), 10875–10900, https://doi.org/10.5194/acp-22-10875-2022, **2022**.
27. Kerr, G. H., **Goldberg, D. L**., Knowland, K. E., Keller, C. A., Oladini, D., Kheirbek, I., Mahoney, L., Lu, Z. and Anenberg, S. C.: Diesel passenger vehicle shares influenced COVID-19 changes in urban nitrogen dioxide pollution, *Environ. Res. Lett*., 17(7), 074010, https://doi.org/10.1088/1748-9326/AC7659, **2022**.
28. Ahn, D. Y., Salawitch, R. J., Canty, T. P., He, H., Ren, X. R., **Goldberg, D. L.,** Dickerson, R. R. The U.S. power sector emissions of CO2 and NOx during 2020: Separating the impact of the COVID-19 lockdowns from the weather and decreasing coal in fuel-mix profile, Atmospheric Environment: X, Volume 14, https://doi.org/10.1016/j.aeaoa.2022.100168, **2022**.
29. Tzortziou, M., Kwong, C. F., **Goldberg, D. L.,** Schiferl, L., Commane, R., Abuhassan, N., Szykman, J. and Valin, L.: Declines and peaks in NO_2 pollution during the multiple waves of the COVID-19 pandemic in the New York metropolitan area, Atmos. Chem. Phys., 1–30, https://doi.org/10.5194/ACP-2021-592, **2022**.
30. Jing, P. and **Goldberg, D. L.:** Influence of conducive weather on ozone in the presence of reduced NOx emissions: A case study in Chicago during the 2020 lockdowns, Atmos. Pollut. Res., 13(2), 101313, https://doi.org/10.1016/J.APR.2021.101313, **2022**.
31. Anenberg, S. C., Mohegh, A., **Goldberg, D. L.,** Kerr, G. H., Brauer, M., Burkart, K., Hystad, P., Larkin, A., Wozniak, S. and Lamsal, L.: Long-term trends in urban NO_2 concentrations and associated paediatric asthma incidence: estimates from global datasets, Lancet Planet. Heal., 6(1), e49–e58, https://doi.org/10.1016/S2542-5196(21)00255-2, **2022**.

2021 (3 first-author, 9 total)

32. **Goldberg, D. L.,** Anenberg, S. C., Lu, Z., Streets, D. G., Lamsal, L., McDuffie, E. and Smith, S.: Urban NOx emissions around the world declined faster than anticipated between 2005 and 2019, *Environ. Res. Lett.,* https://doi.org/10.1088/1748-9326/AC2C34, **2021.**
33. **Goldberg, D. L.,** Anenberg, S. C., Kerr, G. H., Lu, Z. and Streets, D. G.: TROPOMI: A revolutionary new satellite instrument measuring NO_2 air pollution, *Environ. Manag.,* **2021.**

34. **Goldberg, D. L.,** Anenberg, S. C., Mohegh, A., Lu, Z. and Streets, D. G.: TROPOMI NO_2 in the United States: A detailed look at the annual averages, weekly cycles, effects of temperature, and correlation with surface NO_2 concentrations, *Earth's Future,* https://doi.org/10.1029/2020EF001665, **2021.**
35. Nawaz, M. O., Henze, D. K., Harkins, C., Cao, H., Nault, B., Jo, D., Jimenez, J., Anenberg, S. C., **Goldberg, D. L.** and Qu, Z.: Impacts of sectoral, regional, species, and day-specific emissions on air pollution and public health in Washington, DC, Elem. Sci. Anthr., 9(1), https://doi.org/10.1525/elementa.2021.00043, **2021**.
36. Laughner, J. L., …, **Goldberg, D. L.,** …, Zeng, Z.-C.: Societal shifts due to COVID-19 reveal large-scale complexities and feedbacks between atmospheric chemistry and climate change, *Proc. Natl. Acad. Sci.,* 118(46), https://doi.org/10.1073/PNAS.2109481118, **2021.**
37. Anenberg, S., Kerr, G. H. and **Goldberg, D. L**.: Leveraging satellite data to address air pollution inequities, *Environ. Manag.,* **2021.**
38. Kondragunta, S., Wei, Z., McDonald, B. C., **Goldberg, D. L.** and Tong, D. Q.: COVID-19 Induced Fingerprints of a New Normal Urban Air Quality in the United States, J*. Geophys. Res. Atmos*., e2021JD034797, https://doi.org/10.1029/2021JD034797, **2021.**
39. Kerr, G. H., **Goldberg, D. L.** and Anenberg, S. C.: COVID-19 pandemic reveals persistent disparities in nitrogen dioxide pollution, *Proc. Natl. Acad. Sci.,* 118(30), e2022409118, https://doi.org/10.1073/pnas.2022409118, **2021.**
40. Gorris, M. E., Anenberg, S. C., **Goldberg, D. L.,** Kerr, G. H., Stowell, J. D., Tong, D. Q., and Zaitchik, B. F.: Shaping the future of science: COVID-19 highlighting the importance of GeoHealth, *GeoHealth*, 5, e2021GH000412, https://doi.org/10.1029/2021gh000412, **2021**.

<u>2020 (1 first-author, 5 total)</u>

41. **Goldberg, D. L.,** Anenberg, S. C., Griffin, D., McLinden, C. A., Lu, Z. and Streets, D. G.: Disentangling the Impact of the COVID-19 Lockdowns on Urban NO_2 From Natural Variability, *Geophys. Res. Lett.,* 47(17), https://doi.org/10.1029/2020GL089269, **2020.**
42. Mohegh, A., **Goldberg, D. L.,** Achakulwisut, P. and Anenberg, S. C.: Sensitivity of estimated NO_2 -attributable pediatric asthma incidence to grid resolution and urbanicity, *Environ. Res. Lett.,* https://doi.org/10.1088/1748-9326/abce25, **2020.**
43. Anenberg, S. C., Bindl, M., Brauer, M., Castillo, J. J., Cavalieri, S., Duncan, B. N., Fiore, A. M., Fuller, R., **Goldberg, D. L.,** Henze, D. K., Hess, J., Holloway, T., James, P., Jin, X., Kheirbek, I., Kinney, P. L., Liu, Y., Mohegh, A., Patz, J., Jimenez, M. P., Roy, A., Tong, D., Walker, K., Watts, N. and West, J. J.: Using Satellites to Track Indicators of Global Air Pollution and Climate Change Impacts: Lessons Learned From a NASA-Supported Science-Stakeholder Collaborative, *GeoHealth,* 4(7), https://doi.org/10.1029/2020GH000270, **2020.**
44. Saide, P. E., Gao, M., Lu, Z., **Goldberg, D. L.,** Streets, D. G., … , and Crawford, J. H.: Understanding and improving model representation of aerosol optical properties for a Chinese haze event measured during KORUS-AQ, *Atmos. Chem. Phys*., 20(11), 6455–6478, https://doi.org/10.5194/acp-20-6455-2020, **2020.**

45. Liu, F., Duncan, B. N., Krotkov, N. A., Lamsal, L. N., Beirle, S., Griffin, D., McLinden, C. A., **Goldberg, D. L.** and Lu, Z.: A methodology to constrain carbon dioxide emissions from coal-fired power plants using satellite observations of co-emitted nitrogen dioxide, *Atmos. Chem. Phys.,* 20(1), 99–116, https://doi.org/10.5194/acp-20-99-2020, **2020.**

<u>2019 (4 first-author, 4 total)</u>

46. **Goldberg, D. L.,** Saide, P. E., Lamsal, L. N., De Foy, B., Lu, Z., Woo, J.-H., Kim, Y., Kim, J., Gao, M., Carmichael, G. and Streets, D. G.: A top-down assessment using OMI NO_2 suggests an underestimate in the NOx emissions inventory in Seoul, South Korea, during KORUS-AQ, *Atmos. Chem. Phys.,* 19(3), https://doi.org/10.5194/acp-19-1801-2019, **2019.**
47. **Goldberg, D. L.,** Lu, Z., Streets, D. G., de Foy, B., Griffin, D., McLinden, C. A., Lamsal, L. N., Krotkov, N. A. and Eskes, H.: Enhanced Capabilities of TROPOMI NO_2 : Estimating NOx from North American Cities and Power Plants, *Environ. Sci. Technol.,* acs.est.9b04488, https://doi.org/10.1021/acs.est.9b04488, **2019.**
48. **Goldberg, D. L.,** Lu, Z., Oda, T., Lamsal, L. N., Liu, F., Griffin, D., McLinden, C. A., Krotkov, N. A., Duncan, B. N. and Streets, D. G.: Exploiting OMI NO_2 satellite observations to infer fossil-fuel CO_2 emissions from U.S. megacities, Sci. Total Environ., 695, 133805, https://doi.org/10.1016/j.scitotenv.2019.133805, **2019.**
49. **Goldberg, D. L.,** Gupta, P., Wang, K., Jena, C., Zhang, Y., Lu, Z. and Streets, D. G.: Using gap-filled MAIAC AOD and WRF-Chem to estimate daily PM 2.5 concentrations at 1 km resolution in the Eastern United States, *Atmos. Environ.,* 199, 443–452, https://doi.org/10.1016/j.atmosenv.2018.11.049, **2019.**

<u>2018 (0 first-author, 1 total)</u>

50. Ring, A. M., Canty, T. P., Anderson, D. C., Vinciguerra, T. P., He, H., **Goldberg, D. L.,** Ehrman, S. H., Dickerson, R. R. and Salawitch, R. J.: Evaluating commercial marine emissions and their role in air quality policy using observations and the CMAQ model, *Atmos. Environ.,* 173(June 2017), 96–107, https://doi.org/10.1016/j.atmosenv.2017.10.037, **2018.**

<u>2017 (1 first-author, 1 total)</u>

51. **Goldberg, D. L.,** Lamsal, L. N., Loughner, C. P., Swartz, W. H., Lu, Z. and Streets, D. G.: A high-resolution and observationally constrained OMI NO_2 satellite retrieval, *Atmos. Chem. Phys.,* 17(18), 11403–11421, https://doi.org/10.5194/acp-17-11403-2017, **2017.**

<u>2016 (1 first-author, 2 total)</u>

52. **Goldberg, D. L.,** Vinciguerra, T. P., Anderson, D. C., Hembeck, L., Canty, T. P., Ehrman, S. H., Martins, D. K., Stauffer, R. M., Thompson, A. M., Salawitch, R. J. and Dickerson, R. R.: CAMx ozone source attribution in the eastern United States using guidance from observations during DISCOVER-AQ Maryland, *Geophys. Res. Lett.,* 43(5), 2249–2258, https://doi.org/10.1002/2015GL067332, **2016.**
53. Ren, X., Luke, W. T., Kelley, P., Cohen, M. D., Artz, R., Olson, M. L., Schmeltz, D., Puchalski, M., **Goldberg, D. L.,** Ring, A., Mazzuca, G. M., Cummings, K. A., Wojdan, L., Preaux, S. and Stehr, J. W.: Atmospheric mercury measurements at a suburban site in

the Mid-Atlantic United States: Inter-annual, seasonal and diurnal variations and source-receptor relationships, *Atmos. Environ.,* 146, 141–152, https://doi.org/10.1016/j.atmosenv.2016.08.028, **2016.**

<u>2015 (1 first-author, 3 total)</u>

54. **Goldberg, D. L.,** Vinciguerra, T. P., Hosley, K. M., Loughner, C. P., Canty, T. P., Salawitch, R. J. and Dickerson, R. R.: Evidence for an increase in the ozone photochemical lifetime in the eastern United States using a regional air quality model, *J. Geophys. Res. Atmos.,* 120(24), 12778–12793, https://doi.org/10.1002/2015JD023930, **2015.**
55. Canty, T. P., Hembeck, L., Vinciguerra, T. P., Anderson, D. C., **Goldberg, D. L.,** Carpenter, S. F., Allen, D. J., Loughner, C. P., Salawitch, R. J. and Dickerson, R. R.: Ozone and NOx chemistry in the eastern US: Evaluation of CMAQ/CB05 with satellite (OMI) data, *Atmos. Chem. Phys.,* 15(19), 10965–10982, https://doi.org/10.5194/acp-15-10965-2015, **2015**.
56. Stauffer, R. M., Thompson, A. M., Martins, D. K., Clark, R. D., **Goldberg, D. L.,** Loughner, C. P., Delgado, R., Dickerson, R. R., Stehr, J. W. and Tzortziou, M. A.: Bay breeze influence on surface ozone at Edgewood, MD during July 2011, *J. Atmos. Chem.,* 72(3–4), 335–353, https://doi.org/10.1007/s10874-012-9241-6, **2015.**

<u>2014 (1 first-author, 2 total)</u>

57. **Goldberg, D. L.,** Loughner, C. P., Tzortziou, M., Stehr, J. W., Pickering, K. E., Marufu, L. T. and Dickerson, R. R.: Higher surface ozone concentrations over the Chesapeake Bay than over the adjacent land: Observations and models from the DISCOVER-AQ and CBODAQ campaigns, *Atmos. Environ.,* 84, 9–19, https://doi.org/10.1016/j.atmosenv.2013.11.008, **2014.**
58. Loughner, C. P., Tzortziou, M., Follette-Cook, M., Pickering, K. E., **Goldberg, D. L.,** Satam, C., Weinheimer, A., Crawford, J. H., Knapp, D. J., Montzka, D. D., Diskin, G. S. and Dickerson, R. R.: Impact of bay-breeze circulations on surface air quality and boundary layer export, *J. Appl. Meteorol. Climatol.,* 53(7), 1697–1713, https://doi.org/10.1175/JAMC-D-13-0323.1, **2014.**

<u>Manuscripts Under Review (3)</u>

1. Chen, L., **Goldberg, D., L.,** Huber, D. E., Anenberg, S. C., Zhang, Y., Kerr, G. H. Daytime dynamics of NO_2 inequity: new insights from the TEMPO instrument. Submitted.
2. Goldberg, P. M., Abhishek Anand, A., **Goldberg, D. L.**, Westervelt, D M. Variable Short-Term Air Quality Impacts of New York City's Congestion Pricing Policy. Submitted.
3. Lang, V., Partha, D., **Goldberg, D. L.**, Nawaz, M. O., Kerr, G. H., Chen, L., Janssen, M., Adelman, Z., Urbaszewski, B., Deylami, N., Minuche, M., Chavez, G., Horton, D. A framework for augmenting medium- and heavy-duty vehicle emissions within the National Emissions Inventory for hyperlocal air quality assessments. Submitted.

Competitive Grants Selected for Funding (14 of 30 submitted proposals funded; 47%)

1. "Societal benefits of TEMPO NO_2: Applications for air quality management", $400,000, ***Principal Investigator***
2. "Evaluating Updates to CAMx and NOx Emission Inventories using TEMPO Measurements over Texas", September 2024 – September 2025, $230,000, ***Co-Investigator,*** (PI: Jeremiah Johnson, Ramboll)
3. "Efficacy of Vehicle Emission Control Interventions in Ameliorating Air Pollution Exposure and Health Burdens in Marginalized Communities", August 2024 – August 2027, $1,000,000, **Co-Investigator** (PI: Daniel Horton, Northwestern University)
4. "Assessing environmental justice air quality, and health co-benefits of transport interventions in US urban areas", NASA, November 2023 – June 2025, $1,200,000, ***Co-Investigator*** (PI: Gaige Kerr, George Washington University)
5. "Pushing the boundaries of fine-scale NO_X emission quantification from remote sensing instruments", NASA, April 2023– March 2026, $750,000, ***Principal Investigator***
6. "Source-sector NOx emissions analysis with sub-kilometer scale airborne observations in Houston during TRACER-AQ", Air Quality Research Program (AQRP), August 2022– August 2023, $ 248,146, ***Principal Investigator***
7. "Air Quality Modeling to Investigate Environmental Injustice in Washington, DC & Baltimore", George Washington University Research Innovation Award, July 2022– June 2024, $75,000, ***Principal Investigator***
8. "Value of GeoXO atmospheric composition data for estimating air pollution-related health impacts", NOAA, August 2021 – September 2025, $800,000, ***Co-Investigator*** (PI: Susan Anenberg, George Washington University).
9. "Using satellite NO_2 observations for public health surveillance and environmental policy planning at global, national, and urban scales", NASA HAQAST, January 2021 – January 2025, $800,000, ***Co-Investigator*** (PI: Susan Anenberg, George Washington University).
10. "Inconsistent effects of social distancing on air quality in global cities: lessons for protecting near-term public health and designing longer-term urban transportation policies", NASA Rapid Response, June 2020 – May 2021, $100,000, ***Co-Principal Investigator*** (Co-PI: Susan Anenberg, George Washington University).
11. "Updating the Wisconsin Horizontal Interpolation Program for Satellites (WHIPS)", Texas Commission on Environmental Quality Research Program, June 2020 – November 2021, $200,000, ***Co-Investigator*** (PI: Tracey Holloway, University of Wisconsin).
12. "Integrating satellites, ground monitoring, and modeling to estimate long-term NO_2 exposures and associated pediatric asthma impacts", Health Effects Institute, November 2019 – November 2021, $120,000, ***Co-Investigator*** (PI: Susan Anenberg, George Washington University).
13. "The Changing Atmosphere in North America for 2000 – 2020: High-resolution modeling and satellite analysis", NASA ACMAP, May 2019 – April 2023, $750,000, ***Co-Investigator*** (PI: Greg Carmichael, University of Iowa).
14. "Taking OMI NO_2 to the next level: Inferring global fossil fuel CO_2 emissions using OMI NO_2 Data Improved with Critical Algorithm Updates", NASA ACMAP, June 2017 – May 2020, $750,000, ***Co-Investigator*** (PI: Nick Krotkov, NASA Goddard)

Invited Seminars (24 total)

1. **Goldberg, D. L.**, et al. Visualizing Air Pollution from Space. Presented at University of Maryland – College Park. College Park, MD. **Sept 2026.**
2. **Goldberg, D. L.**, et al. Visualizing Air Pollution from Space. Presented at University of Texas – San Antonio. San Antonio, TX. **Feb 2026.**
3. **Goldberg, D. L.**, et al. Visualizing Air Pollution from Space. Presented at Arizona State University. Tempe, AZ. **Feb 2026.**
4. **Goldberg, D. L.**, et al. Satellite monitoring of air pollution. Presented at the Earth Justice Monthly Seminar. Washington, DC. **July 2025.**
5. **Goldberg, D. L.**, et al. The many successes of satellite NO_2. Presented at the Air & Waste Management (AWMA) Webinar. **May 2025.**
6. **Goldberg, D. L.**, et al. The many successes of satellite NO_2. Presented at the Air & Waste Management (AWMA) Chesapeake Bay Regional Webinar. **April 2025.**
7. **Goldberg, D. L.,** et al. Visualizing Air Pollution from Space. Presented at SXSW. Austin, TX. **Mar 2024.**
8. **Goldberg, D. L.**, et al. Policy- and health-relevant applications of NO_2 satellite measurements. Presented at the SESS-NOAA Air Composition Meeting. Fairfax, VA. **Feb 2024.**
9. **Goldberg, D. L.**, et al. Policy and Health Relevant Applications of TROPOMI NO_2 in Urban Areas. Presented at the MDE Photochemical Modeling and Data Analysis Seminar Series. **March 2023.**
10. **Goldberg, D. L.,** et al. Applications of satellite data to understand the concentrations and emission rates of NO_2 air pollution. Presented at the City College of New York (CCNY), Earth and Atmospheric Sciences Weekly Seminar Series. New York City, NY. **Nov 2022.**
11. **Goldberg, D. L.**, et al. Pushing the methodological limits on satellite use in health applications. Presented at the HEI Health Applications for Satellite-Derived Air Quality Workshop. **May 2022.**
12. **Goldberg, D. L.**, et al. Applications of satellite remote sensing to better understand global air quality. Presented at Rutgers University. **April 2022.**
13. **Goldberg, D. L.**, et al. Elucidating the “new normal” of NOx emissions in urban areas. Presented at Boston University. **Feb 2022.**
14. **Goldberg, D. L.**, et al. Using NO_2 satellite data for urban environmental justice applications and lessons learned during the COVID-19 lockdowns. Presented at the New York State BAQAR/NYSERDA Seminar. **Jan 2022.**
15. **Goldberg, D. L.**, et al. Estimating NOx emissions from cities using satellite data. Presented at the WESTAR Council Annual Meeting. Santa Fe, New Mexico. **Dec 2021.**
16. **Goldberg, D. L.**, et al. Estimating NOx emissions from global cities using satellite data. Presented at the NCAR ACOM Weekly Seminar. **April 2021.**
17. **Goldberg, D. L.**, et al. Health impacts of NO_2 and its changes during the COVID-19 lockdowns. NCAR Climate and Health Seminar Series. **Sept 2020.**
18. **Goldberg, D. L.**, et al. Using satellite data to estimate air pollution at high spatiotemporal resolution. Presented at The Applications for Big Data and the Environment. Davis, CA. **Jan 2020.**

19. **Goldberg, D. L.**, et al. Using NASA satellite data to estimate exposure to air pollution. Presented at the George Washington University Environmental and Occupational Health Seminar. Washington, DC. **Dec 2019.**
20. **Goldberg, D. L.**, et al. Estimating air pollution emissions, exposures, and public health impacts in cities worldwide. Presented at the New Applications in the Use of Satellite Data Monitoring for Population Health. Huntsville, AL. **Oct 2019.**
21. **Goldberg, D. L.**, et al. Using satellite data to estimate air pollution at high spatiotemporal resolution. Presented at The Workshop in Environmental Economics and Data Science (TWEEDS). Portland, OR. **March 2019.**
22. **Goldberg, D. L.**, et al. Innovative techniques to observe and model air pollution in the eastern United States. Presented at Northwestern University. Evanston, IL. **Aug 2017.**
23. **Goldberg, D. L.**, et al. Innovative techniques to observe and model air pollution in the eastern United States. Presented at the University of Wisconsin-Madison. Madison, WI. **June 2017.**
24. **Goldberg, D. L.**, et al. Lifetime and distribution of ozone air pollution in the eastern United States. Presented at Carnegie Mellon University. Pittsburgh, PA. **Feb 2016.**

<u>First-Author Conference & Science Team Presentations (57 total)</u>

1. **Goldberg, D. L.**, et al. Advancing Surface NO_2 and NOx Emission Estimates through TEMPO and Model Integration. Presented at the NASA TEMPO Science Team Meeting. Iowa City, IA. **Jun 2026.**
2. **Goldberg, D. L.**, et al. Contrasting the TEMPO and TROPOMI Views of Tropospheric NO_2. Presented at the AIRDC Annual Meeting. Fairfax, VA. **Jun 2026.**
3. **Goldberg, D. L.**, et al. Contrasting the TEMPO and TROPOMI Views of Tropospheric NO_2. NASA HAQAST Spring Meeting. Madison, WI. **May 2026.**
4. **Goldberg, D. L.**, et al. Contrasting the TEMPO and TROPOMI Views of Tropospheric NO_2: Implications for Diurnal Variability and Model Evaluation. TEMPO Monthly Webinar. **Feb 2026.**
5. **Goldberg, D. L.**, et al. Connecting TEMPO Column NO_2 to Surface NO_2 and NO_x Emission Patterns. Presented at the NASA TEMPO Science Team Meeting. Boston, MA. **Aug 2025.**
6. **Goldberg, D. L.**, et al. Connecting TEMPO Column NO_2 to Surface NO_2 and NO_x Emission Patterns. Presented at the DC Researchers Consortium Air Quality Meeting. Baltimore, MD. **Jun 2025.**
7. **Goldberg, D. L.**, et al. Connecting TEMPO Column NO_2 to Surface NO_2 and NO_x Emission Patterns. Presented at the NASA TEMPO Science Team Teleconference. **Apr 2025.**
8. **Goldberg, D. L.**, et al. The many successes of satellite NO_2. Presented at the NASA HAQAST Final Showcase. Washington, DC. **Jan 2025.**
9. **Goldberg, D. L.**, et al. Policy and health relevant applications of NO_2 remote sensing data. Presented at the AGU Fall Meeting. Washington, DC. **Dec 2024.**
10. **Goldberg, D. L.**, et al. How does the clear-sky bias of satellite measurements affect the estimates of annualized near-surface NO_2? Presented at the AGU Fall Meeting. Washington, DC. **Dec 2024.**

11. **Goldberg, D. L.**, et al. Source-sector NOx emissions analysis with sub-kilometer scale airborne observations in Houston during TRACER-AQ. Presented at the MARAMA Annual Meeting. Richmond, VA. **Nov 2024.**
12. **Goldberg, D. L.**, et al. Applications of OMI and TROPOMI NO_2 data for Public Health at Urban Scales. Presented at the HAQAST Spring Meeting. Boston, MA. **June 2024.**
13. **Goldberg, D. L.**, et al. Source-sector NOx emissions analysis with sub-kilometer scale airborne observations in Houston during TRACER-AQ. Presented at the TRACER-AQ Science Team Meeting. Houston, TX. **Apr 2024.**
14. **Goldberg, D. L.**, et al. Investigating Hyper-Local NOx emissions in Houston and New York City using GCAS, TROPOMI and Sub-kilometer Model Simulations. Presented at the American Meteorological Society Annual Meeting. Baltimore, MD. **Jan 2024.**
15. **Goldberg, D. L.**, et al. Evaluating the trends and spatial patterns of NOx emissions in urban areas using TROPOMI. Presented at the HAQAST Fall Meeting. Salt Lake City, UT. **Oct 2023.**
16. **Goldberg, D. L.**, et al. Source-sector NOx emissions analysis with sub-kilometer scale airborne observations in Houston during TRACER-AQ. Presented at the TRACER-AQ Science Team meeting. **Oct 2023.**
17. **Goldberg, D. L.**, et al. How will TEMPO data improve our current research? Understanding NOx emissions for environmental health applications. Presented at the TEMPO Science Team Meeting. Huntsville, AL. **May 2023.**
18. **Goldberg, D. L.**, et al. Applications of TROPOMI NO_2 to understand urban NO_2 and NO_X emissions. Presented at the AGU Fall Meeting. Chicago, IL. **Dec 2022.**
19. **Goldberg, D. L.**, et al. Policy and Health Relevant Applications of TROPOMI NO_2. Presented at the HAQAST Fall Meeting. Madison, WI. **Oct 2022.**
20. **Goldberg, D. L.**, et al. Policy and Health Relevant Applications of TROPOMI NO_2 in the United States. Presented at the Sentinel 5-Precursor 5th Anniversary Conference. Taormina, Italy. **Oct 2022.**
21. **Goldberg, D. L.**, et al. Policy and Health Relevant Applications of TROPOMI NO_2: Preparing for TEMPO. Presented at the HAQAST Summer Meeting. Houston, TX. **June 2022.**
22. **Goldberg, D. L.**, et al. Using NO_2 satellite data for urban planning. Presented at the HAQAST Winter 2022 Meeting. **Jan 2022.**
23. **Goldberg, D. L.**, et al. Reconciling differences between satellite-inferred NOx emissions and inventories in global cities. Presented at the AGU Fall Meeting. New Orleans, LA. **Dec 2021.**
24. **Goldberg, D. L.**, et al. Satellite-derived NOx emissions for 80 global megacities between 2005 and 2019. Presented at the IGAC Biannual Meeting. **Sept 2021.**
25. **Goldberg, D. L.**, et al. Changes in NO_2 following the COVID-19 lockdowns: Disentangling anthropogenic changes from natural variability. Presented at the MARAMA Mobile Sources Workshop. **March 2021.**
26. **Goldberg, D. L.**, et al. Top-down NOx emissions estimates for 50 global cities during the last 15 years. Presented at the AMS Annual Meeting. **Jan 2021.**

27. **Goldberg, D. L.**, et al. Using TROPOMI and re-analysis meteorology to disentangle the impact of the COVID-19 lockdowns on urban NO_2 natural variability. Presented at the AGU Fall Meeting. **Dec 2020.**
28. **Goldberg, D. L.**, et al. Changes in NO_2 following the COVID-19 lockdowns: Disentangling anthropogenic changes from natural variability. Presented at the MARAMA Science Team Meeting. **Dec 2020.**
29. **Goldberg, D. L.**, et al. Health impacts of NO_2 and its changes during the COVID-19 lockdowns. Co-presentation at the OTC Annual Meeting. **Nov 2020.**
30. **Goldberg, D. L.**, et al. Oversampling TROPOMI NO_2 in the US & using it to estimate effects of COVID-19 lockdowns on urban NOx emissions. Presented at the OMI-TROPOMI Science Team Meeting. **Oct 2020**.
31. **Goldberg, D. L.**, et al. Disentangling the impact of the COVID-19 lockdowns on urban NO_2 from natural variability. Presented at the 19th Annual CMAS Conference. **Oct 2020.**
32. **Goldberg, D. L.**, et al. The satellite data revolution: How new satellite instruments can provide better estimates of NOx pollution. Presented at the ISEE Annual Meeting. **Aug 2020.**
33. **Goldberg, D. L.**, et al. Estimating air pollution emissions, exposures, and public health impacts in cities worldwide. Presented at the TOLNET and Pandora Science Team Meeting. **June 2020.**
34. **Goldberg, D. L.**, et al. Using models and satellite data to estimate air pollution in the Baltimore-Washington area. Presented at The Baltimore-Washington Regional Air Quality Symposium. College Park, MD. **Feb 2020.**
35. **Goldberg, D. L.**, et al. Estimating air pollution emissions, exposures, and public health impacts in cities worldwide. Presented at the American Meteorological Society Annual Meeting, Boston, MA. **Jan 2020.**
36. **Goldberg, D. L.**, et al. Using OMI NO_2 to infer fossil-fuel emissions of CO_2 from large metropolitan areas in the United States. Presented at the Aura Science Team Meeting. Pasadena, CA. **Aug 2019.**
37. **Goldberg, D. L.**, et al. Policy-relevant applications of satellite data: Estimating air pollution emissions, exposures, and public health impacts in cities worldwide. Presented at the HAQAST Summer 2019 Meeting. Pasadena, CA. **July 2019.**
38. **Goldberg, D. L.**, et al. Investigating NO_X emissions from megacities using re-processed OMI NO_2 and TROPOMI NO_2. Presented at the OWLETS Science Team Meeting. College Park, MD. **May 2019.**
39. **Goldberg, D. L.**, et al. Recent Advances in Deriving NOx Emission Estimates from Satellite Data. Presented at the AGU Fall Meeting. Washington, D.C. **Dec 2018.**
40. **Goldberg, D. L.**, et al. Linking Surface Monitors, Satellite Data, and Emissions Inventories to Investigate Regional Haze Trends in the Eastern U.S. Presented at the American Geophysical Union (AGU) Fall Meeting. Washington, D.C. **Dec 2018.**
41. **Goldberg, D. L.**, et al. Using MAIAC AOD and WRF-Chem to estimate daily $PM_{2.5}$ concentrations at 1 km resolution in the eastern United States. Presented at the CMAS Conference. Chapel Hill, NC. **Oct 2018.**

42. **Goldberg, D. L.**, et al. A top-down assessment using OMI NO_2 suggests an underestimate in the NO_x emissions inventory in Seoul, Korea during KORUS-AQ. Presented at the KORUS-AQ Science Team Meeting. Irvine, CA. **Aug 2018.**
43. **Goldberg, D. L.**, et al. Recent advances in estimating NO_x emissions from OMI. Presented at the HAQAST Summer 2018 Meeting. Madison, WI. **July 2018.**
44. **Goldberg, D. L.**, et al. Estimating NO_x emissions and surface concentrations at high spatial resolution. Presented at the AGU Fall Meeting. New Orleans, LA. **Dec 2017.**
45. **Goldberg, D. L.**, et al. Estimating NO_x emissions and surface concentrations at high spatial resolution. Presented at the HAQAST Fall 2017 Meeting. Palisades, NY. **Nov 2017.**
46. **Goldberg, D. L.**, et al. Ground measurements, satellite observations, and model simulations of air quality in the Chesapeake Bay region. Presented at the OWLETS Science Team Meeting. Baltimore, MD. **Oct 2017.**
47. **Goldberg, D. L.**, et al. A new satellite technique to derive high-resolution tropospheric NO_2 columns in the eastern United State. Presented at the OMI Science Team Meeting. Greenbelt, MD. **Sept 2017.**
48. **Goldberg, D. L.**, et al. Validation of a satellite technique to derive high-resolution tropospheric NO_2 columns in Korea. Presented at the KORUS-AQ Science Team Meeting. Seogwipo, Jeju, South Korea. **Feb 2017.**
49. **Goldberg, D. L.**, et al. High-resolution OMI satellite retrievals of tropospheric NO_2 in the eastern United States. Presented at the 15th Annual CMAS Conference. Chapel Hill, NC. **Oct 2016.**
50. **Goldberg, D. L.**, et al. Evidence for an increasing geographic region of influence on ozone air pollution in the eastern United States. Presented at the CMAS Conference. Chapel Hill, NC. **Oct 2015.**
51. **Goldberg, D. L.**, et al. Recent ozone modeling results in the Mid-Atlantic. Presented at the MARAMA Science Team Meeting. Richmond, VA. **July 2015.**
52. **Goldberg, D. L.**, et al. Evidence for an increasing geographic region of influence on ozone air pollution in the Eastern United States. Presented at the NASA AQAST Summer 2015 Meeting. St Louis, MO. **June 2015.**
53. **Goldberg, D. L.**, et al. Scientific insight from CAMx OSAT modeling. Presented at the OTC Spring Meeting. Washington, DC. **April 2015.**
54. **Goldberg, D. L.**, et al. The Impact of the Chesapeake Bay Climate and Boundary Layer Dynamics on Air Pollutant Concentrations during Smog Episodes. Presented at the AMS Annual Meeting. Phoenix, AZ. **Jan 2015.**
55. **Goldberg, D. L.**, et al. Recent Improvements in Regional Air Quality Models and their Impacts on Ozone Source Attribution. Presented at the CMAS Conference. Chapel Hill, NC. **Oct 2014.**
56. **Goldberg, D. L.**, et al. Increased Air Pollution over the Chesapeake Bay and its Effect on Deposition to the Bay. Presented at the NADP Fall Meeting. Indianapolis, IN. **Oct 2014.**
57. **Goldberg, D. L.**, et al. Higher surface ozone concentrations over the Chesapeake Bay than over adjacent land: Observations and models from DISCOVER-AQ. Presented at the DISCOVER-AQ Science Team Meeting. Newport News, VA. **Feb 2014.**

First-Author Conference Poster Presentations (10 total)

1. **Goldberg, D. L.**, et al. Evaluating the spatial patterns of NOx emissions in polluted areas with TROPOMI NO_2. Presented at the TEMPO Science Team Meeting. Huntsville, AL. **May 2023**
2. **Goldberg, D. L.**, et al. High-resolution NO_2 exposure estimates and top-down NO_X emissions using OMI NO_2 and TROPOMI NO_2. Presented at the AGU Fall Meeting. San Francisco, CA. **Dec 2019**
3. **Goldberg, D. L.**, et al. Using MAIAC AOD to estimate daily PM2.5 and its long-term trends (2008 – 2018) at 1 km resolution in the Eastern United States. Presented at the EPA ACE Annual Meeting. Pittsburgh, PA. **June 2019**
4. **Goldberg, D. L.**, et al. Using MAIAC AOD to estimate daily PM2.5 and its long-term trends (2008 – 2018) at 1 km resolution in the Eastern United States. Presented at the OWLETS Science Team Meeting. College Park, MD. **May 2019**
5. **Goldberg, D. L.**, et al. Using MODIS AOD and WRF-Chem to infer daily PM2.5 concentrations at 1 km resolution in the eastern United States. Presented at the HAQAST Summer 2018 Meeting. Madison, WI. **July 2018**
6. **Goldberg, D. L.**, et al. High resolution satellite retrievals of NO_2 and Aerosol Optical Depth for health impact studies. Presented at the AGU Fall 2016 Meeting. San Francisco, CA. **Dec 2016**
7. **Goldberg, D. L.**, et al. CAMx Ozone Source Attribution in the Eastern United Sates using Guidance from Observations during DISCOVER-AQ Maryland. Presented at the NASA AQAST Winter 2016 Meeting. RTP, NC. **Jan 2016**
8. **Goldberg, D. L.**, et al. Using Source Apportionment to Evaluate the Cross State Transport of Ozone in the Eastern United States. Presented at the AGU Fall Meeting. San Francisco, CA. **Dec 2014**
9. **Goldberg, D. L.**, et al. Using CAMx and CMAQ to Investigate Cross-state Transport of Ozone in the Eastern United States. Presented at the NASA AQAST Summer 2014 Meeting. Boston, MA. **June 2014**
10. **Goldberg, D. L.**, et al. CAMx and CMAQ Model Intercomparison for July 2007 in the Baltimore-Washington Metropolitan Region. Presented at the AGU Fall Meeting. San Francisco, CA. **Dec 2013**

Mentoring: Junior Scientists Advised

Post-doc (7 total): Dr. Arash Mohegh (2019 – 2021), Dr. Gaige Kerr (2020 – 2022), Dr. Doyeon Ahn (2021 – 2024), Dr. Katelyn O'Dell (2021 – 2023), Dr. Tess Carter (2022 – 2023), Dr. Omar Nawaz (2023 – 2025), Dr. Daniel Huber (2023 – 2025), Dr. Lulu Chen (2024 – present)

Post-Bachelors (7 total): Nigel Martis (2021 – 2022), Xavier Nogueira (2021 – 2022), Perrin Krisko (2021 – 2023), Rishabh Khanna (2022), Ziyad Maknojia (2022), Sara Runkel (2023 – 2024), Erin Campbell (2024 – 2025)

Undergraduate (3 total): Duncan Quevedo (2021), Emily Richardt (2022), Olivia Paquette (2024)

Teaching Experience

2023 – 2026 George Washington University, Sustainable Energy and Environmental Health
3-credit undergraduate class for upperclassmen (30 students)

2020 – 2022 George Washington University, Global Climate Change and Air Pollution
Guest Lecture on using measurements and models to estimate air pollution

2012 – 2013 University of Maryland, AOSC200: Introduction to Weather and Climate
Head TA. Taught 2 lectures per week. Graded all exams, quizzes, and projects.

Service & Leadership

2026	Faculty Mentor	NASA Student Airborne Research Program (SARP)
2026	Expert Reviewer	NASA Tropospheric Composition Report
2019–2026	Panel Reviewer	NASA ROSES
2024	Expert Reviewer	EPA NO_x Integrated Science Assessment
2024	Lead Discussant	EPA Ozone NAAQS Pre-Review Workshop
2021–2024	Technical Advisory	Metro Washington DC Air Quality Committee
2019–2022	Panel Reviewer	NASA Postdoctoral Program (NPP)
2021	Panel Reviewer	National Science Foundation (NSF)
2021	Panel Reviewer	Netherlands Organization for Scientific Research
2021	Mentor	NASA DEVELOP Internship Program
2021	Session Organizer	American Meteorological Society
2019–2020	Session Organizer	American Geophysical Union
2016–2018	Executive Board Member	Argonne Postdoctoral Society
2016–2017	Mentor	ACT-SO High School Science Competition
2012–2014	Site Operator	Clean Air Status & Trends Network (CASTNET)
2012–2014	Site Operator	National Atmospheric Deposition Program (NADP)

Media & Public Engagement

1. WTOP, July 17, 2026, Unhealthy air in DC region with smoky skies from Canadian wildfires, https://wtop.com/weather-news/2026/07/dc-region-under-code-purple-air-quality-alert-due-to-heat-and-smoky-skies/
2. New York Times, March 25, 2026, Subject Matter Expert for Journalists Mira Rojanasakul, Hiroko Tabuchi, and Harry Stevens
3. WTOP, March 8, 2024, Here are the worst areas for air pollution in DC, according to satellite imagery, https://wtop.com/local/2024/03/here-are-the-worst-areas-for-air-pollution-in-dc-according-to-satellite-imagery/
4. WUSA9, February 23, 2023, Did toxic plumes from an Ohio train derailment impact DC?, https://www.wusa9.com/article/news/verify/verify-derailed-train-toxic-plumes-did-not-reach-dc-area/65-13b9b112-4bc5-45b5-9147-bc1e6c7990e1
5. WUSA9, February 24, 2022, Humans are driving climate change but they can also fight against it, https://www.wusa9.com/article/tech/science/environment/dnp-yes-humans-are-contributing-to-climate-change-global-warming-greenhouse-gas-emissions/65-8a6e9b6a-faa2-4560-80a3-cfcfbadf56f5
6. NASA Earth Observatory, October 28, 2021, Scientific Questions Arrive in Ports, https://earthobservatory.nasa.gov/images/149004/scientific-questions-arrive-in-ports

7. NBC Connecticut, July 26, 2021, A Look at How Western Wildfire Smoke Makes Its Way to Connecticut, http://nbcct.co/Czz0Ari
8. New York Times, April 15, 2021, Subject Matter Expert for Journalist Lisa Collins
9. Washington Post, June 16, 2020, Commentary in, “Washington has yet to see unhealthful pollution levels this year. That’s a record”, https://www.washingtonpost.com/weather/2020/06/16/washington-dc-record-low-pollution/
10. WAMU 88.5, NPR Radio, May 19, 2020, Subject Matter Expert for The Kojo Nnamdi Show, https://thekojonnamdishow.org/audio/#/shows/2020-05-19/reducing-air-pollution-during-the-pandemic/116753/@00:00
11. Washington Post, April 22, 2020, Commentary in, “Washington has its cleanest spring air in 25 years: How air quality has improved during the coronavirus crisis”, https://www.washingtonpost.com/weather/2020/04/22/washington-dc-air-quality-coronavirus/
12. Nature, April 10, 2020, Commentary in, “Why pollution is plummeting in some cities — but not others”, https://www.nature.com/articles/d41586-020-01049-6
13. New York Times, April 3, 2020, Subject Matter Expert for Visual Investigations Journalist Christoph Koettl
14. WTOP Radio, April 1, 2020, Commentary in, “Despite telework, stay-at-home orders, not much change to air quality in DC area”, https://wtop.com/local/2020/04/despite-telework-stay-at-home-orders-not-much-change-to-air-quality-in-dc-area/
15. Science Magazine, February 12, 2020, Commentary in, “Deadly air pollution is blowing into your state from a surprisingly large source”, https://www.sciencemag.org/news/2020/02/deadly-air-pollution-blowing-your-state-surprisingly-large-source
16. Washington Post, February 3, 2020, Commentary in, “Why a toxic brown haze loomed over the Capitol on Monday”, https://www.washingtonpost.com/weather/2020/02/03/how-toxic-brown-haze-loomed-over-capitol-monday/
17. Washington Post, July 5, 2019, Commentary in, “Lost in a wall of smoke: Why so many people couldn’t see Washington’s Fourth of July fireworks”, https://www.washingtonpost.com/weather/2019/07/05/lost-wall-smoke-why-so-many-people-couldnt-see-washingtons-july-th-fireworks/
18. Washington Post, February 4, 2019, Commentary in, “There’s an air quality alert in Washington, the sky is hazy and it’s February. What’s going on?”, https://www.washingtonpost.com/weather/2019/02/04/theres-an-air-quality-alert-washington-sky-is-hazy-its-february-whats-going/
19. Washington Post, July 10, 2018, Commentary in, “Washington posted first Code Red day since 2012 on Monday due to ‘unhealthy’ pollution levels”, https://www.washingtonpost.com/news/capital-weather-gang/wp/2018/07/10/washington-posted-first-code-red-day-since-2012-monday-due-to-unhealthy-pollution-levels/

Awards & Honors

2020	Impact Argonne Award for Extraordinary Effort
2015	Outstanding Graduate Assistant Award for top 2% of all graduate students
2012	Service Award for dedicated service to UMD Atmospheric Science Dept
2009	American Chemical Society (ACS) Award for top chemical engineer in class
2008	Tau Beta Pi Engineering Honor Society for top 20% of engineering class

Professional Memberships

2011 – 2026 American Geophysical Union (AGU)
2011 – 2026 American Meteorological Society (AMS)